

DATABASE_SCHEMA

Database Schema

Management approach

Mechanism	Role
Hibernate ddl-auto=update	Creates/updates most tables from entities
schema.sql	Creates account_ordinal_seq
Seed runners	RoleInitializer, AdminRoleInitializer, AccountPolicyInitializer

Entity-relationship (current)

erDiagram

```
customers ||--o{ account : has
users ||--o{ user_roles : has
roles ||--o{ user_roles : grants
account_policy {
    bigint policy_id PK
    string account_type
    string currency_code
    boolean opening_deposit_required
    decimal required_opening_deposit
    decimal minimum_balance
    boolean active
}
customers {
    bigint customer_id PK
    string first_name
    string last_name
    string email UK
    string phone_number UK
    date date_of_birth
    string address
    string status
    timestamp created_at
    timestamp updated_at
}
account {
    bigint account_id PK
    string account_number UK
    string branch_code
    string account_type
    int ordinal
    string currency_code
    int check_digit
    decimal available_balance
    decimal ledger_balance
    int debit_count
    int credit_count
    string status
```

```

    bigint customer_id FK
}
users {
    bigint user_id PK
    string username UK
    string password_hash
    boolean enabled
    int failed_attempts
    boolean locked
}
roles {
    bigint role_id PK
    string name UK
}
user_roles {
    bigint user_role_id PK
    bigint user_id FK
    bigint role_id FK
}

```

Tables

customers

Entity: `com.bankone.customer.entity.Customer`

Column	Notes
<code>customer_id</code>	PK, identity
<code>first_name, last_name</code>	Required
<code>email, phone_number</code>	Unique
<code>date_of_birth</code>	Optional
<code>address, status</code>	Required
<code>created_at, updated_at</code>	@PrePersist / @PreUpdate

`customerCode` is **not** stored; JSON accessor formats ID via `BusinessIdFormatter`.

account

Entity: `com.bankone.account.entity.Account`

Column	Notes
<code>account_id</code>	PK
<code>account_number</code>	Unique, generated
<code>branch_code</code>	String (no branch table)
<code>account_type, currency_code, ordinal, check_digit</code>	Number composition
<code>available_balance, ledger_balance</code>	Updated on open/deposit
<code>debit_count, credit_count, last*_at</code>	Counters / stamps
<code>status</code>	ACTIVE, FROZEN, DORMANT, SUSPENDED, CLOSED
<code>customer_id</code>	FK → customers
Audit-ish	<code>created_at, activated_at, closed_at, created_by, closed_by</code>

account_policy

Entity: `com.bankone.account.entity.AccountPolicy` extends `AuditableEntity`

Unique (`account_type`, `currency_code`).

Seeded INR policies for CURRENT, SAVINGS, SALARY, FIXED_DEPOSIT, RECURRING_DEPOSIT. **No LOAN seed.**

users, roles, user_roles

Entities under `com.bankone.user` / `com.bankone.role`. Sequences: `user_seq`, `role_seq`, `user_role_seq` (Hibernate-managed naming may vary by dialect config).

`AuditableEntity` columns on User/Role/UserRole/AccountPolicy: `created_at`, `updated_at`, `created_by`, `updated_by`, `version`.

Sequences

```
-- schema.sql
CREATE SEQUENCE IF NOT EXISTS account_ordinal_seq
START WITH 1 INCREMENT BY 1;
```

Used by `AccountRepository.getNextOrdinal()`.

Not present (planned)

No tables yet for: `transaction` / `ledger entries`, `branch`, `loan product`, `audit_event`, `beneficiary`.

Migration notes

Changing entity fields with `ddl-auto=update` alters tables in place. For production-ready history, replace with Flyway/Liquibase (see `EXTENSION_GUIDE`).